

To: Director, School of Engineering (Kwok Siong Teh, PhD.)

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Subject: Funding Request for Engr 696-01: Open-Source Battery Assessment and Management System (OSBAMS) — Rev.2 Rebuild

1. Problem Statement and Background

OSBAMS is an individual capstone project developing a low-cost, open-source lithium-ion battery pack assessment platform. The current prototype (Rev.1) validated the core control and sensing architecture, but two problems must be resolved before the system can be considered a complete, safe, standalone unit.

First, fit and containment: the power-stage electronics (E-stop, battery disconnect switch, relay, current shunt, and control PCB) are housed in an enclosure, but the Raspberry Pi and touchscreen display currently have no enclosure of their own, and the system as a whole does not yet fit together as a single standalone unit.

Second, a specific hardware issue was identified during Rev.1 bring-up that requires a PCB and component rebuild rather than a patch: the TC74 temperature sensor is currently non-functional (offline fault) and requires a new interface circuit. The current/voltage sensing (INA228, rated to 85 V, above the pack's 42 V fully-charged voltage) is already implemented and working, but is soldered directly to the Rev.1 board; the Rev.2 board spin requires its own populated INA228 rather than desoldering and reusing the existing one.

Background: the design builds on standard open-source battery-management and EV disconnect-switching practices (fused, E-stop-protected DC power paths with independent hardware interlocks), which informed both the original sensor selection and the disconnect-switching component discussed below.

2. Proposed Solution and Design

Rev.2 addresses both problems with a coordinated rebuild:

Enclosure overhaul: a single, properly sized enclosure to house the Raspberry Pi, touchscreen, wiring harnesses, and power electronics together as one standalone, safely contained unit.

PCB redesign: a corrected board spin carrying over the working INA228 current/voltage sensing (already validated on Rev.1) and repairing the TC74 temperature interface.

Main disconnect upgrade: the main battery disconnect switching element was originally specified as a genuine Albright SW60 contactor, selected for its rating for switching battery/traction loads. Time and budget constraints during the initial prototype build made this infeasible, and a lower-cost automotive relay (Durakool, 80A) was substituted. Rev.2 restores the original design intent: a genuine SW60 with auxiliary contact, sourced directly through Albright's exclusive US distributor, Curtis Instruments.

Figure 1 shows the current Rev.1 power-stage wiring, which Rev.2 carries forward into the unified enclosure.

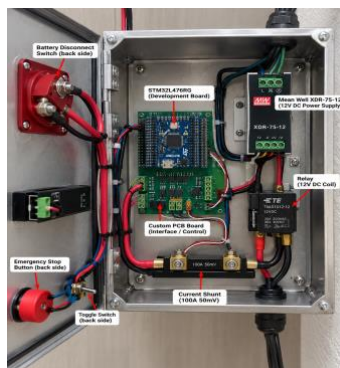


Figure 1. Rev.1 power-stage wiring w labeled.

Figure 2 is an AI-rendered concept illustration of the intended Rev.2 layout, not a photo of built hardware, shown to communicate planned component placement.

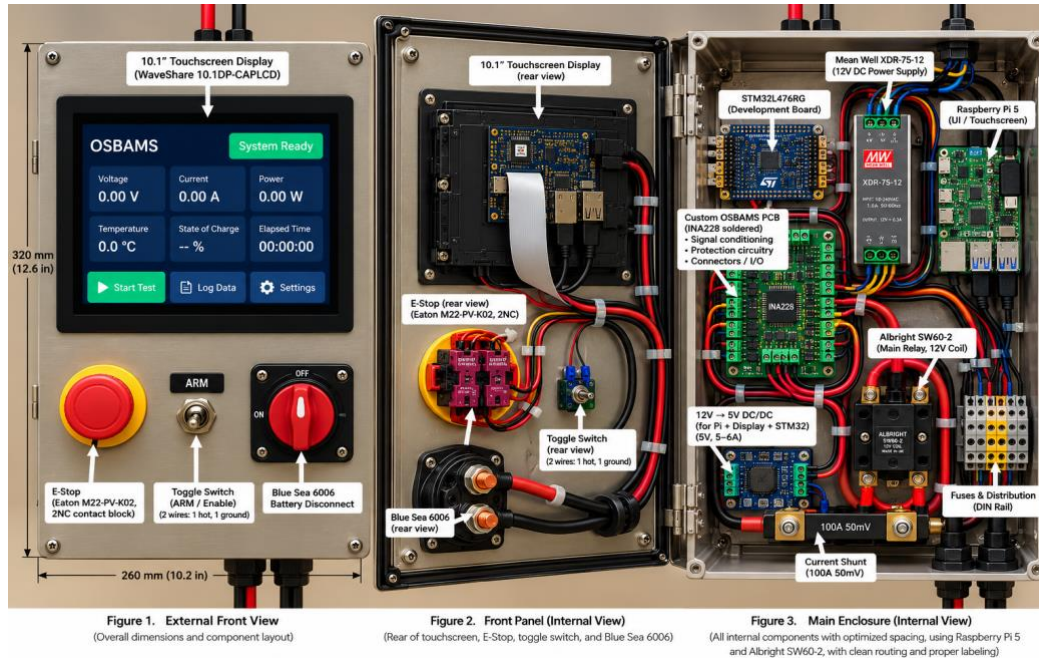


Figure 2. Conceptual Rev.2 layout (AI-rendered illustration, not a photo of built hardware).

3. Bill of Materials (BOM) & Cost Summary

Table 1. OSBAMS Rev.2 Bill of Materials.

Material	Pre-tax	Tax-incl.
Rev.2 custom PCB fabrication, 4 boards (incl. est. 15% China-fab tariff)	\$115.00	\$124.92
PCB component set (incl. est. 12% import tariff)	\$78.40	\$85.16
Spare PCB components/rework parts (incl. est. 12% import tariff)	\$39.20	\$42.58
Spare NUCLEO-L476RG (incl. est. 12% import tariff)	\$33.60	\$36.50
INA228/spare sensing hardware (incl. est. 12% import tariff)	\$28.00	\$30.41
RSA-20-50 shunt/sensing hardware (incl. est. 12% import tariff)	\$28.00	\$30.41
TC74 + ADC protection/divider parts (incl. est. 12% import tariff)	\$28.00	\$30.41
Genuine Albright SW60 + auxiliary contact (Curtis Instruments, custom order; incl. est. 10% UK-origin tariff)	\$165.00	\$179.23
E-stop + wiring/safety hardware	\$45.00	\$48.88
Fuse holders + fuses/spares (incl. est. 12% import tariff)	\$39.20	\$42.58
12V→5V DC/DC converter (Raspberry Pi 5 + touchscreen + STM32, powered from XDR-75-12; incl. est. 12% import tariff)	\$50.40	\$54.75
Raspberry Pi 5, 8GB (real MSRP, Feb 2026 memory-price increase)	\$125.00	\$135.78

Material	Pre-tax	Tax-incl.
WaveShare 10.1" touchscreen (Micro Center, SKU 967059, Santa Clara pickup)	\$95.99	\$105.35
microSD/cables/adapters	\$20.00	\$21.72
Final enclosure, 24x16x12in steel IP65 (real listing; Pi + touchscreen + power stage)	\$151.11	\$164.14
DIN rails/end stops/clips	\$28.00	\$30.41
DIN terminal blocks/distribution	\$30.00	\$32.59
Cable glands/strain relief	\$20.00	\$21.72
Standoffs/brackets/display mounting	\$24.00	\$26.07
Fans/vents/filter materials	\$20.00	\$21.72
Battery connector/adaptor materials	\$46.00	\$49.97
Wire/ferrules/lugs/terminals	\$50.00	\$54.31
Heat shrink/sleeving/labels	\$24.00	\$26.07
Drilling/step-bit consumables	\$24.00	\$26.07
Cutting/deburring consumables	\$24.00	\$26.07
Rework/soldering/fabrication materials	\$35.00	\$38.02
Total Requested Funding	\$1,366.90	\$1,485.84

4. Budget Justification

The requested total of \$1,485.84 (inclusive of applicable sales tax by vendor location — 8.625% CA/SF for most lines, 9.75% Santa Clara County for the Micro Center touchscreen — and estimated import tariffs on non-US-manufactured components, per current distributor invoicing practice) is tied directly to the two problems above:

PCBs and core components: multiple board spins and a spares allotment are budgeted to prevent a project delay from a single-spin fabrication error, and to populate a new INA228 (currently soldered to the Rev.1 board) and repair the TC74 interface on the new board.

Main disconnect: the genuine Albright SW60 with auxiliary contact restores my original design specification, downscoped in Rev.1 for cost and time reasons. The price shown is an estimate, including a 10% allowance for the current US import tariff on UK-origin goods (Albright is UK-manufactured), pending a formal quote from Curtis Instruments, Albright's US distributor, whom I will contact directly to place a custom order.

Enclosure and DIN hardware: covers the structural enclosure, terminal blocks, DIN rails, and cable management needed to move from a partially-housed prototype to a single, fully enclosed, standalone system.

Display, computing, and consumables: covers the replacement Raspberry Pi, touchscreen, and mounting hardware for the dashboard, plus wiring hardware, ferrules, drill/step bits, and soldering materials for final assembly and wiring distribution.

All requested funds will be spent within Fall 2026. This semester (Engr 696) covers purchasing, the PCB rebuild, enclosure fabrication, assembly and integration, and initial system testing. Engr 697 continues with calibration, validation, and refinement of the completed Rev.2 system.